



Lab 3

Basic Logic Gate Implementation

Part 3

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Objectives

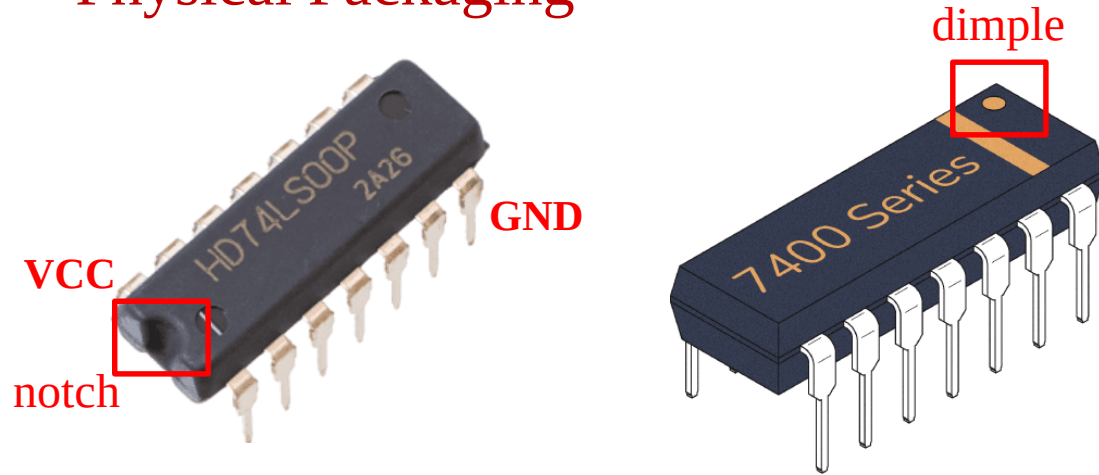
- Understand following components
 - Transistor-Transistor Logic (TTL) ICs 7400, 7404, 7408, 7432
- Implement **logic circuits** to verify the corresponding **truth tables**

TTL 7400 (NAND)

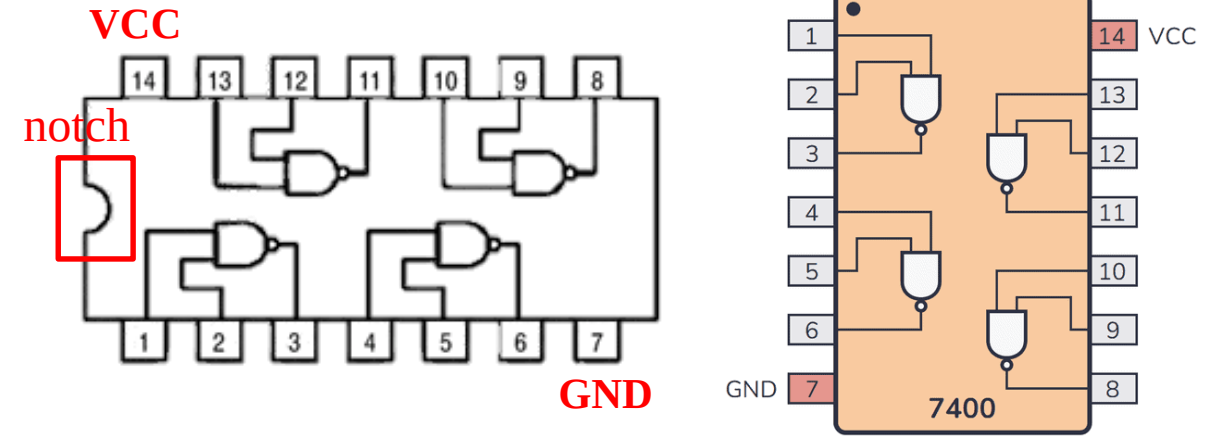
Figure Sources:

- <https://www.igeelectronics.com/products/g69b66b32e/1726810000000161177>
- <https://www.build-electronic-circuits.com/7400-series-integrated-circuits/74hc00-74ls00/>

Physical Packaging



Internal Structure



Graphic Symbol



Truth Table

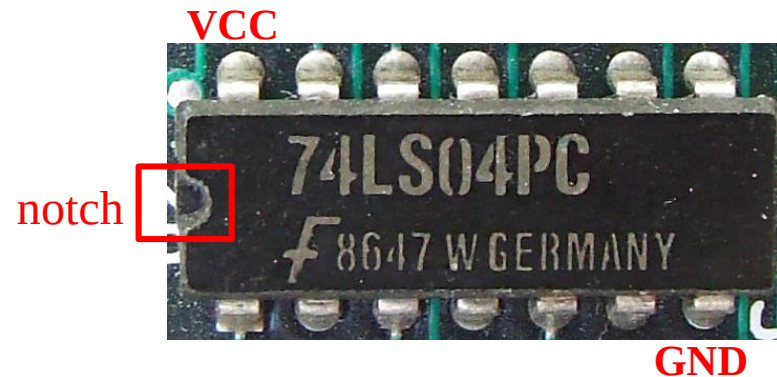
Input A	Input B	Output OUT
0	0	1
0	1	1
1	0	1
1	1	0

TTL 7404 (NOT)

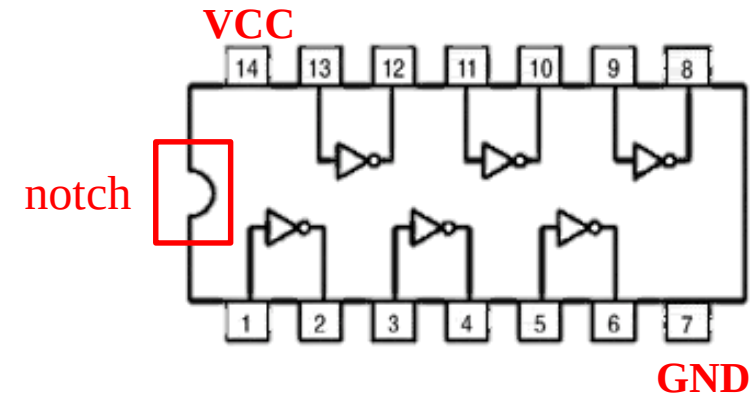
Figure Source:

1. https://pt.wikipedia.org/wiki/TTL_7404

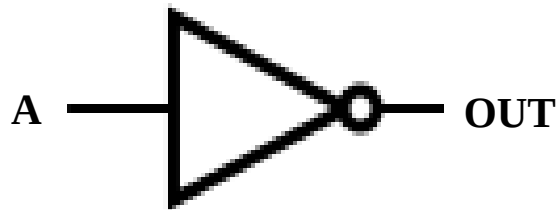
Physical Packaging



Internal Structure



Graphic Symbol



Truth Table

Input A	Output OUT
0	1
1	0

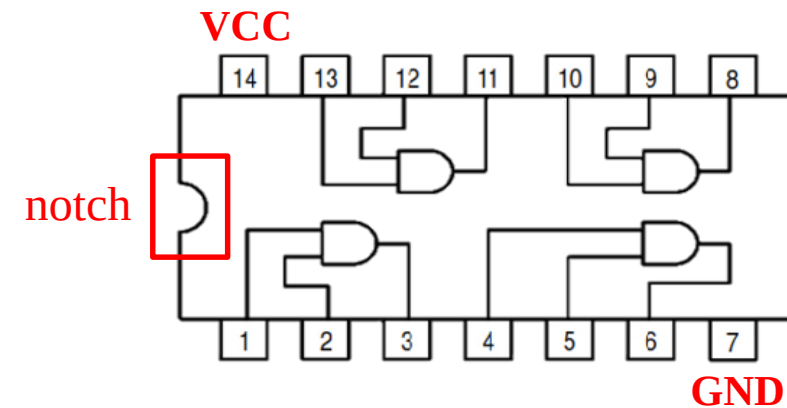
TTL 7408 (AND)

Figure Source:
1. https://pt.wikipedia.org/wiki/TTL_7408

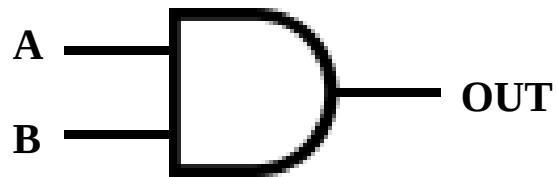
Physical Packaging



Internal Structure



Graphic Symbol



Truth Table

Input A	Input B	Output OUT
0	0	0
0	1	0
1	0	0
1	1	1

TTL 7432 (OR)

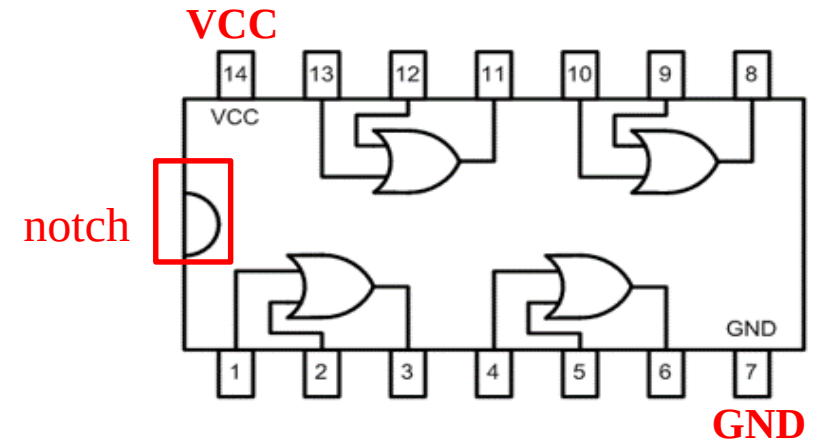
Figure Source:

<https://docs.cirkitdesigner.com/component/e39cedd2-1d38-4573-829f-9f133c89a3b8/7432-or-gate-ic>

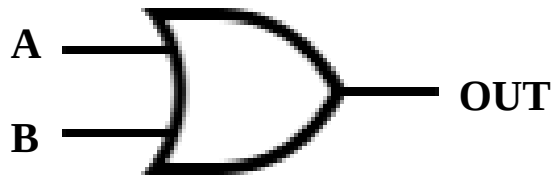
Physical Packaging



Internal Structure



Graphic Symbol



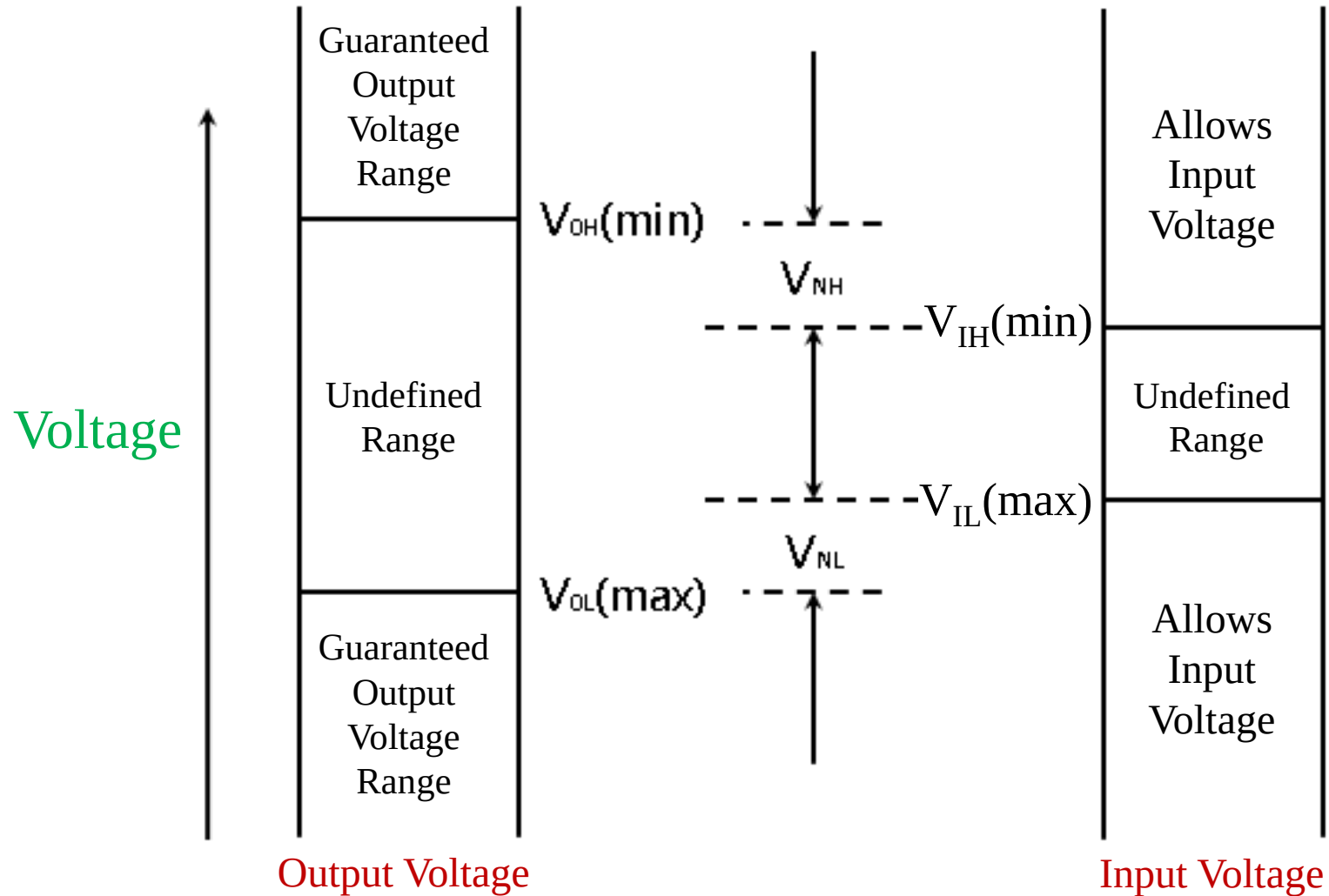
Truth Table

Input A	Input B	Output OUT
0	0	0
0	1	1
1	0	1
1	1	1

Number/Alphabet vs Logic State

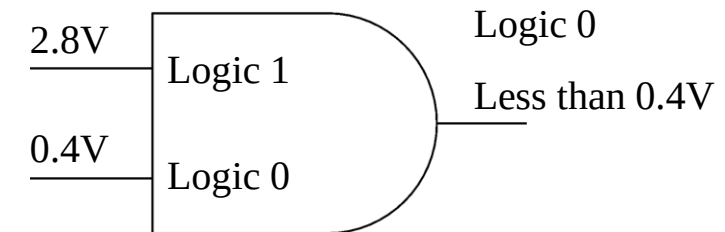
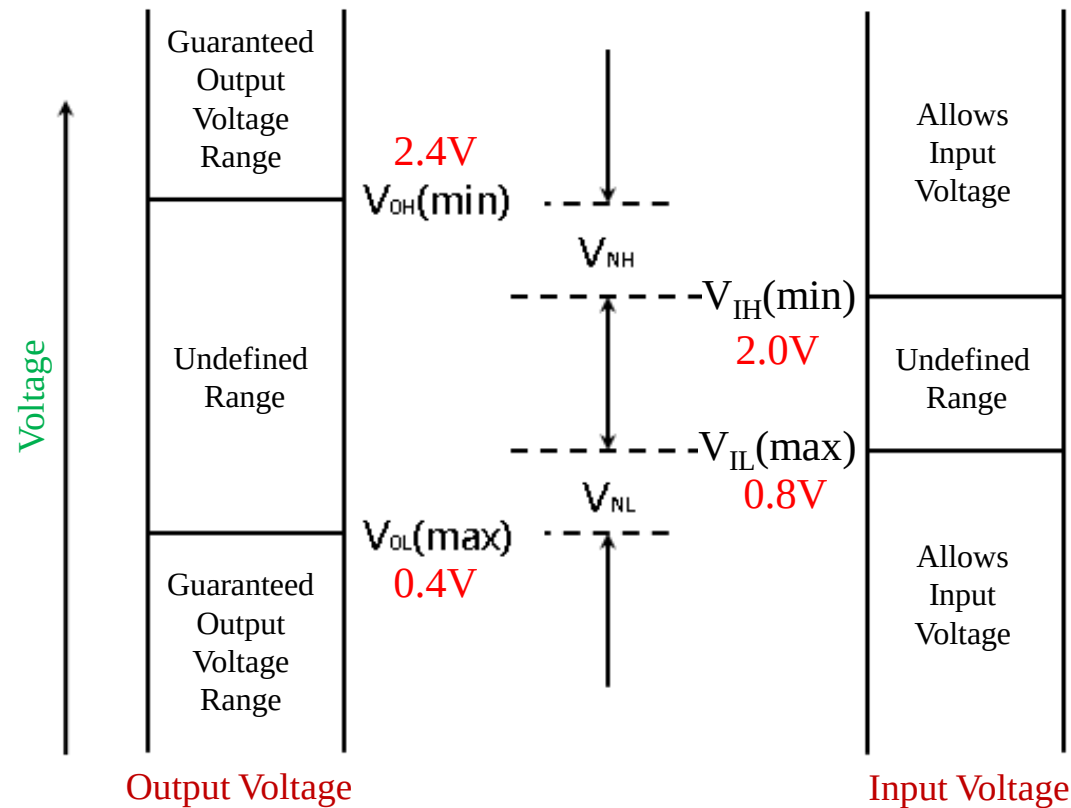
- 0 → Logic LOW (Ground)
- 1 → Logic HIGH (5V)
- x or X → Undefined
- z or Z → High Impedence or Floating
- When an input pin of a TTL IC is floating, it is typically interpreted as a logic HIGH (1).

Voltage Logic Levels



Example of Voltage Logic Levels --- Standard TTL

TTL Type	IC Number	$V_{OH}(\text{min})$	$V_{IH}(\text{min})$	$V_{OL}(\text{max})$	$V_{IL}(\text{max})$	Propagation Delay	Power Consumption
Standard	74xx	2.4V	2.0V	0.4V	0.8V	9 ns	10 mW



Comparison Table

TTL Type	IC Number	$V_{OH}(\text{min})$	$V_{IH}(\text{min})$	$V_{OL}(\text{max})$	$V_{IL}(\text{max})$	Propagation Delay	Power Consumption
Standard	74xx	2.4V	2.0V	0.4V	0.8V	9 ns	10 mW
Low Power	74Lxx	2.4V	2.0V	0.3V	0.8V	33 ns	1 mW
High-Speed	74Hxx	2.4V	2.0V	0.4V	0.8V	6 ns	23 mW
Schottky (high-speed with the same power consumption)	74Sxx	2.4V	2.0V	0.5V	0.8V	3 ns	23 mW
Low Power Schottky	74LSxx	2.4V	2.0V	0.5V	0.8V	9.5 ns	2 mW

Example: Voting System (1/3)

- A teacher (T) and two students (A , B) vote.
 - If the teacher agrees, it passes;
 - If the teacher disagrees, both students must agree for it to pass.

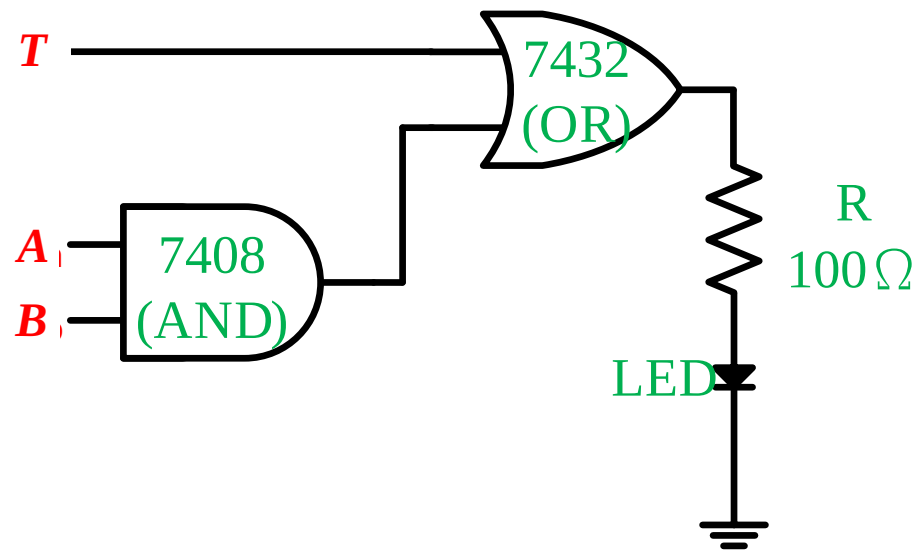
$T \backslash AB$	00	01	11	10
0	0	0	1	0
1	1	1	1	1

$$\text{OUT} = T + A \cdot B$$

Truth Table			
T	A	B	OUT
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

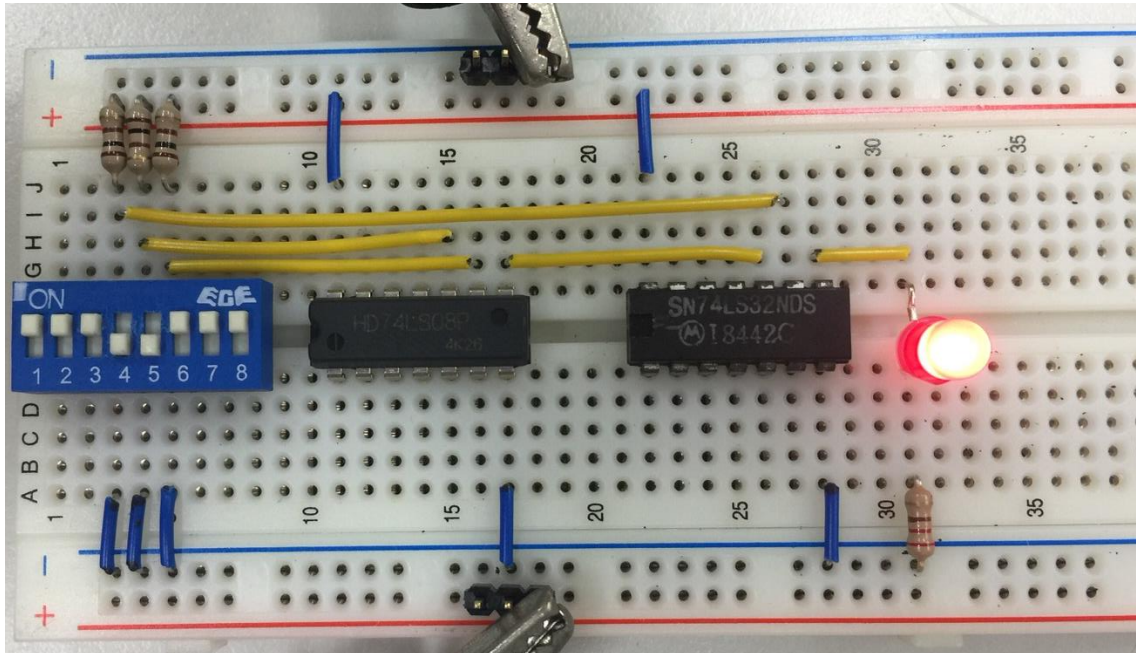
Example: Voting System (2/3)

- The example uses a 7432 (OR gate) and 7408 (AND gate) circuit for practical implementation.



Truth Table				
T	A	B	$A \cdot B$	OUT
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Example: Voting System (3/3)



Truth Table (Logic)

T	A	B	$A \cdot B$	OUT
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Truth Table (Voltage)

T	A	B	$A \cdot B$	OUT
0v	0v	0v	0.16v	0.14v
0v	0v	4.95v	0.16v	0.14v
0v	4.95v	0v	0.16v	0.14v
0v	4.95v	4.95v	4.28v	2.92v
4.95v	0v	0v	0.16v	2.90v
4.95v	0v	4.95v	0.16v	2.91v
4.95v	4.95v	0v	0.16v	2.91v
4.95v	4.95v	4.95v	4.31v	2.93v

LAB 3-1

- Teacher (T) and three students (A, B, C) vote. The result (Out) will pass if either of the following two conditions is met:
 - The teacher and at least one of the students agrees.
 - All three students agree.

- Requirements
 - Draw the truth tables for both logical and voltage.
 - Provide the K-map you used to find the simplest Boolean expression.
 - Based on the simplest Boolean expression, implement the logic circuit on the breadboard with output connected to an LED and take a photo of your implementation.
 - Photo the output of each possible input pattern.

LAB 3-2

- In Keelung, the raining every December is with regularity. Suppose A is Monday, B is Tuesday, C is Wednesday, D is Thursday, and E is Friday. The pattern is as follows:
 - If A , B , C , and D all don't rain, E doesn't rain.
 - If any one of A , B , and C doesn't rain and D doesn't rain, E doesn't rain.
 - If D rains but A , B , and C don't rain, E doesn't rain.
 - If none of the above three conditions apply, E will rain.
 - Based on the above conditions, predict the result for Friday (E).
 - Assume Raining = 1, No raining = 0
- Requirements
 - Draw the truth tables for both logical and voltage.
 - Provide the K-map you used to find the simplest Boolean expression.
 - Based on the simplest Boolean expression, implement the logic circuit on the breadboard with output connected to an LED and take a photo of your implementation.
 - Photo the output of each possible input pattern.

LAB 3-3

- Please follow the logic function below to implement the logic circuit.

$$F = (A \oplus B) + (A'C + BC)$$

- Requirements
 - Draw the truth tables for both logical and voltage.
 - Provide the K-map you used to find the simplest Boolean expression.
 - Based on the simplest Boolean expression, implement the logic circuit on the breadboard with output connected to an LED and take a photo of your implementation.
 - Photo the output of each possible input pattern.

Bonus

- Input: A, B, C, D ; Output: $F(A, B, C, D) = \Pi(1, 2, 5, 7, 8, 9, 10, 11, 13, 15)$
- Requirements
 - Draw the truth tables for both logical and voltage.
 - Provide the K-map you used to find the simplest Boolean expression.
 - Based on the simplest Boolean expression, implement the logic circuit on the breadboard with output connected to an LED.